GlucoTwin system architecture

A per-patient digital twin for Type 2 Diabetes that fuses EHR and wearable data to forecast glucose and warn of post-meal spikes up to 2 hours ahead

1 Data sources

static + dynamic streams

Static EHR (real)

Demographics, BMI
HbA1c, fasting glucose, insulin
Lipid panel (TG, LDL, HDL)
CGMacros bio.csv (PhysioNet)

Synthetic EHR layer (synthetic)

Diagnoses (ICD-10), medications
Family history, BP
TCF7L2 rs7903146, polygenic risk
HbA1c history (-24 to -6 months)
Conditioned on the real labs
Exported as FHIR R4 (ABDM-ready)
Shown in the dashboard record

Wearables / IoT (real)

Dexcom G6 CGM, every 5 min
FreeStyle Libre Pro, every 15 min
Fitbit Sense: heart rate, METs
45 participants, ~10 days each

Meal log (real)

Time, carbs, protein, fat, fibre
Breakfast / lunch / dinner

2 Ingest and align

one clock for every stream

Time alignment

Shared 5-min grid
Gaps <= 15 min interpolated
Unit fix: HbA1c in %
Sentinel values removed
HOMA-IR computed

Meal responses

Baseline, peak, rise, iAUC
Crossed 180 mg/dL?
Clean = full CGM coverage
and no overlapping meal

Live CGM stream

A new reading every 5 min
keeps the running twin
in sync with the patient

labs

meals

3 Digital twin core

physiology, calibrated per patient

EHR prior (day 0)

Ridge regression: labs → twin parameters
A new patient gets a twin from the
health record alone

MAP calibration (days 1-7)

CGM meal curves pull parameters
away from the prior
Personal theta = { S, SG, ka, f }
S = insulin disposition, SG = self-clearance

Physiological model

Gut: Q1 → Q2 (meal absorption, ka)
Ra = f ka Q2 / (VG BW)
dG/dt = -SG E(t)(G - Gb) - X G + Ra
dX/dt = -p2 X + p2 S (G - Gb)+
E(t) = 1 + alpha (METs - 1) exercise
Fat + protein slow absorption (beta)

Data assimilation

Every 5 min the CGM reading nudges
the twin's state (gain 0.6), so each
simulation starts from reality

4 Hybrid forecaster

physics + machine learning

Feature builder (time t)

CGM level and trend
Heart rate, activity
Meals: carbs on board, timing
Labs + twin parameters
Twin's simulated next 2 h

Residual regressors

LightGBM, one per horizon
forecast = twin + correction
+30 / +60 / +120 min

Spike classifier

LightGBM: P(glucose > 180
mg/dL within 2 h)
Threshold set on training patients

SHAP explainer

Top reasons behind
each alert

5 Clinician

decision support

Doctor dashboard

Patients ranked by risk
Live twin + forecast
Spike alert with reasons
What-if simulator:
Indian meal library,
portion, post-meal walk
Glucose profile (AGP), TIR
FHIR health record
Static web app: the twin
runs in the browser

Alert

Risk >= threshold
→ clinician reviews
Decision support only:
never auto-prescribes

real data
synthetic, anchored on real labs
physics twin
machine learning

Evaluation protocol

Unseen patients
5-fold split by patient

Unseen days
only each patient's held-out last 3 days are scored

Baselines
no change, twin only, ML only

Clinical accuracy
Clarke Error Grid zones A-E

Alerts
detection rate, lead time, false alarms per day

Ablations
what each data stream adds, synthetic fields included